

AI/ML Internship – Day 2 Task

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1. OBJECTIVE

The objective of Day 2 was to set up the AI/ML development environment, install required tools and libraries, research computer vision concepts, and implement a basic real-time pose detection system using Python, OpenCV, and MediaPipe.

2. PYTHON INSTALLATION

Step 1 — Python Installation

Python was installed as the primary programming language for AI and Machine Learning development.

Installation Steps:

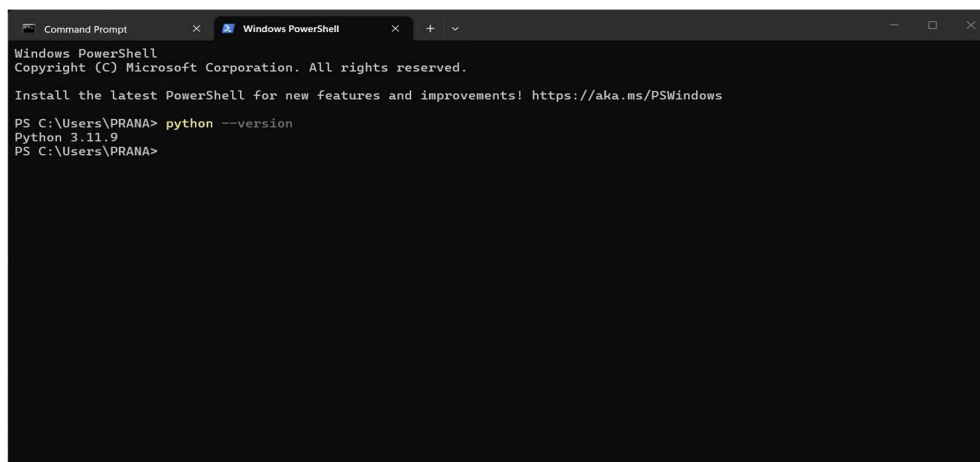
1. Downloaded Python from the official Python website.
2. Installed Python 3.11.
3. Enabled “Add Python to PATH”.
4. Verified installation using Command Prompt.

Verification Command:

```
python --version
```

Example Output:

Python 3.11.9



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\PRANA> python --version
Python 3.11.9
PS C:\Users\PRANA>
```

VS CODE INSTALLATION

Step 2 — VS Code Installation and Setup

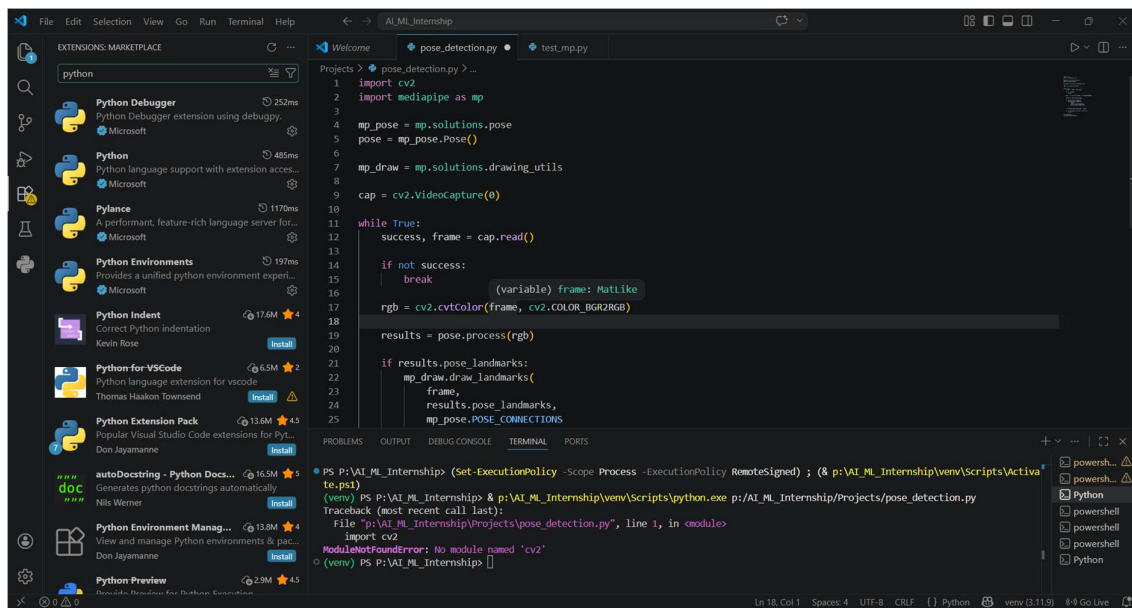
Visual Studio Code was installed as the development environment for writing and executing Python programs.

The following extensions were installed:

- Python Extension
- Pylance
- Python Debugger

These extensions help in:

- code execution
- syntax highlighting
- debugging
- Python environment management



INSTALLATION OF AI/ML LIBRARIES

Step 3 — Installation of Required Libraries

The required AI/ML and computer vision libraries were installed using pip.

Installed Libraries:

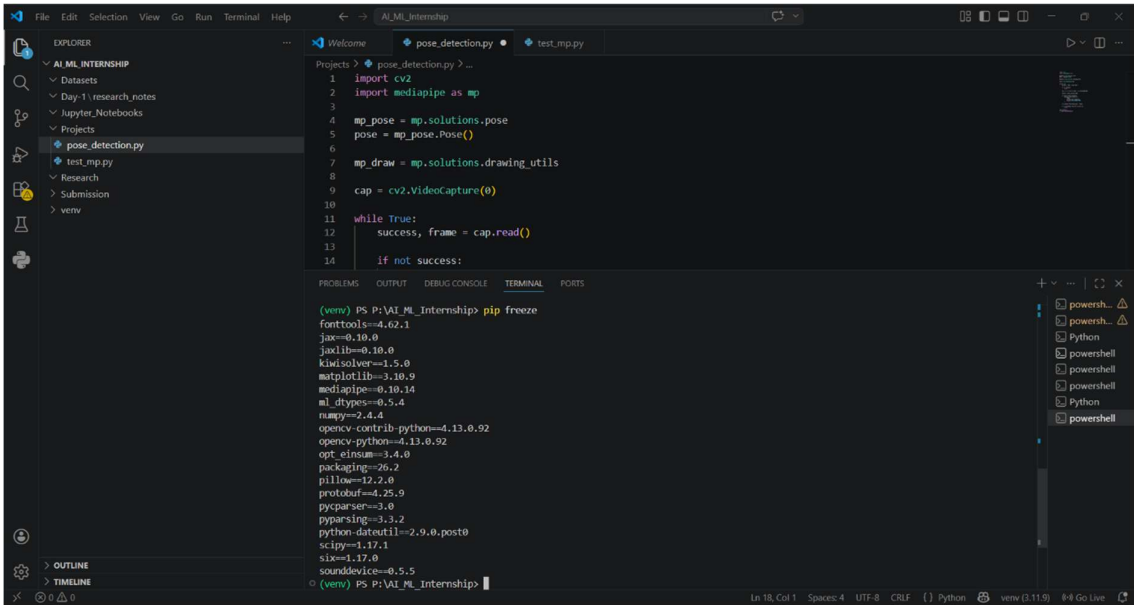
- TensorFlow
- OpenCV
- MediaPipe
- Jupyter Notebook

Installation Commands Used:

pip install tensorflow
pip install opencv-python
pip install mediapipe
pip install notebook

Purpose of Libraries:

Library	Purpose
TensorFlow	Deep learning framework
OpenCV	Image and video processing
MediaPipe	Pose detection and tracking
Jupyter Notebook	Interactive coding environment



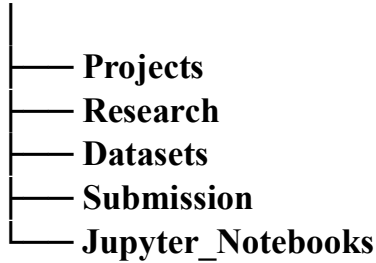
PROJECT FOLDER STRUCTURE

Step 4 — Project Workspace Organization

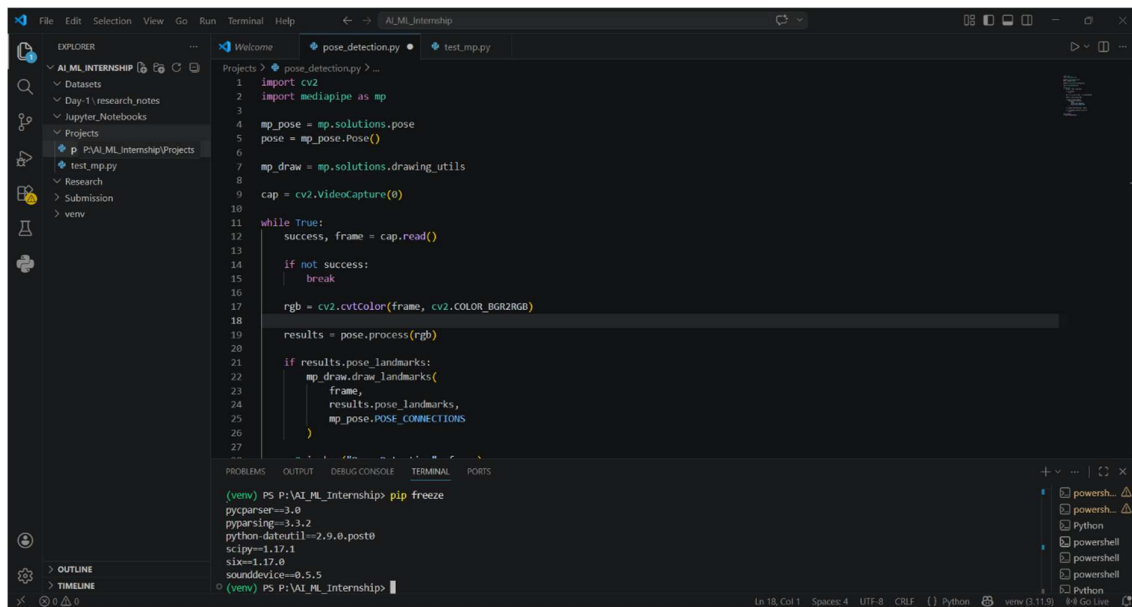
A structured AI/ML project workspace was created in the P drive to organize internship tasks and avoid storage issues in the system drive.

Project Structure:

AI_ML_Internship/



This structure helps maintain project organization and scalability.



RESEARCH TOPICS

1. Body Measurement AI

Body Measurement AI is a computer vision and artificial intelligence technology used to estimate human body measurements using images, videos, or depth sensors instead of traditional manual measuring methods.

The system analyses body shape, posture, and proportions using machine learning algorithms and calculates measurements such as:

- height
- shoulder width
- waist size
- chest size
- hip measurements
- inseam length

This technology is widely used in:

- online fashion shopping
- virtual fitting rooms
- fitness and health applications
- tailoring and apparel industries
- custom clothing platforms

Traditional clothing shopping often results in incorrect size selection because customers cannot physically try products before purchasing. Body Measurement AI solves this issue by generating accurate body measurements digitally.

2. Pose Detection Systems

Pose detection is a computer vision technique that identifies and tracks human body landmarks from images or live video streams.

The system detects important body points such as:

- head
- shoulders
- elbows

- wrists
- hips
- knees
- ankles

Using these landmarks, AI systems can understand:

- posture
- body movement
- gestures
- physical activity

Pose detection systems work by analyzing frames captured from a camera and applying machine learning models to estimate body coordinates in real time.

Technologies commonly used:

- OpenCV
- MediaPipe
- TensorFlow
- Deep Learning Models
- Neural Networks

Applications of pose detection include:

- fitness and exercise tracking
- yoga posture correction
- sports performance analysis
- gaming and motion control
- healthcare rehabilitation
- surveillance systems
- sign language recognition
- virtual try-on systems

In this internship task, MediaPipe and OpenCV were used to implement real-time pose detection using a webcam. The system successfully detected body landmarks and tracked movement dynamically.

3. Virtual Try-On Technology

Virtual Try-On technology is an AI-powered system that allows users to digitally try clothing, accessories, glasses, shoes, or makeup products using cameras and augmented reality.

- The technology combines:
- artificial intelligence
- augmented reality (AR)
- computer vision
- body tracking
- pose estimation

The system first detects the user's body position and measurements, then overlays virtual clothing or products on the user in real time.

- Virtual try-on technology is commonly used in:
- online fashion stores
- eyewear applications
- makeup applications
- footwear shopping platforms
- e-commerce websites

Main advantages:

- improves customer shopping experience
- reduces uncertainty during online purchases
- increases customer confidence
- decreases product return rates
- provides personalized recommendations

The technology helps users visualize how products will appear before purchasing them. Virtual try-on systems became highly popular in the fashion and e-commerce industry because they bridge the gap between physical and online shopping experiences.

IMPLEMENTATION

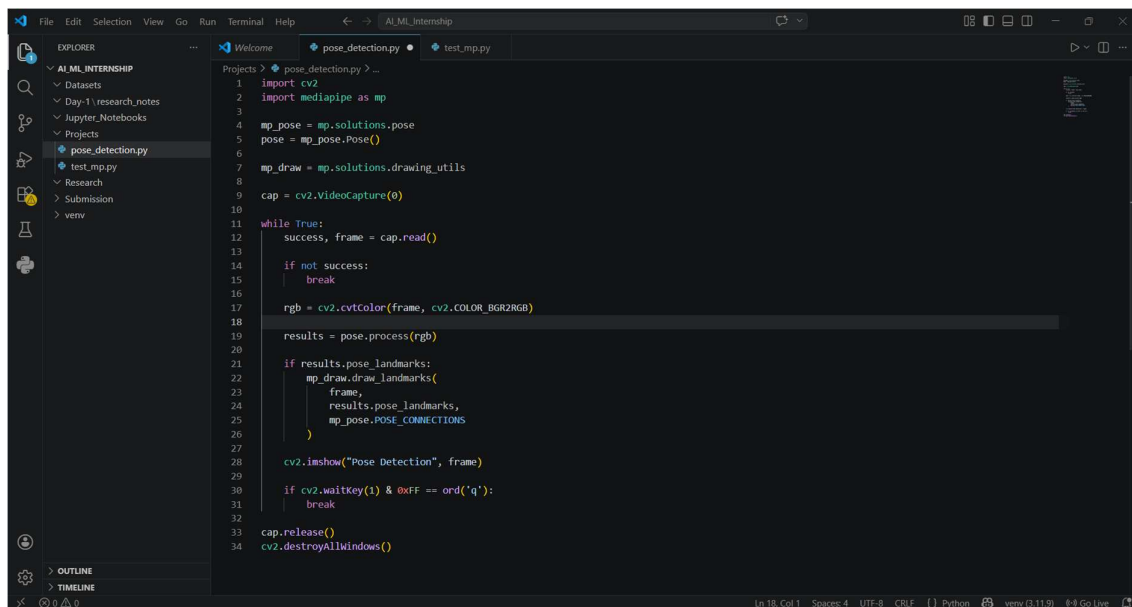
Step 5 — Real-Time Pose Detection Implementation

A real-time pose detection system was implemented using:

- Python
- OpenCV
- MediaPipe

The webcam captured live video input, and MediaPipe detected body landmarks in real time.

The pose detection system successfully tracked body posture and movement dynamically.



```
1 import cv2
2 import mediapipe as mp
3
4 mp_pose = mp.solutions.pose
5 pose = mp_pose.Pose()
6
7 mp_draw = mp.solutions.drawing_utils
8
9 cap = cv2.VideoCapture(0)
10
11 while True:
12     success, frame = cap.read()
13
14     if not success:
15         break
16
17     rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
18
19     results = pose.process(rgb)
20
21     if results.pose_landmarks:
22         mp_draw.draw_landmarks(
23             frame,
24             results.pose_landmarks,
25             mp_pose.POSE_CONNECTIONS
26         )
27
28     cv2.imshow("Pose Detection", frame)
29
30     if cv2.waitKey(1) & 0xFF == ord('q'):
31         break
32
33 cap.release()
34 cv2.destroyAllWindows()
```

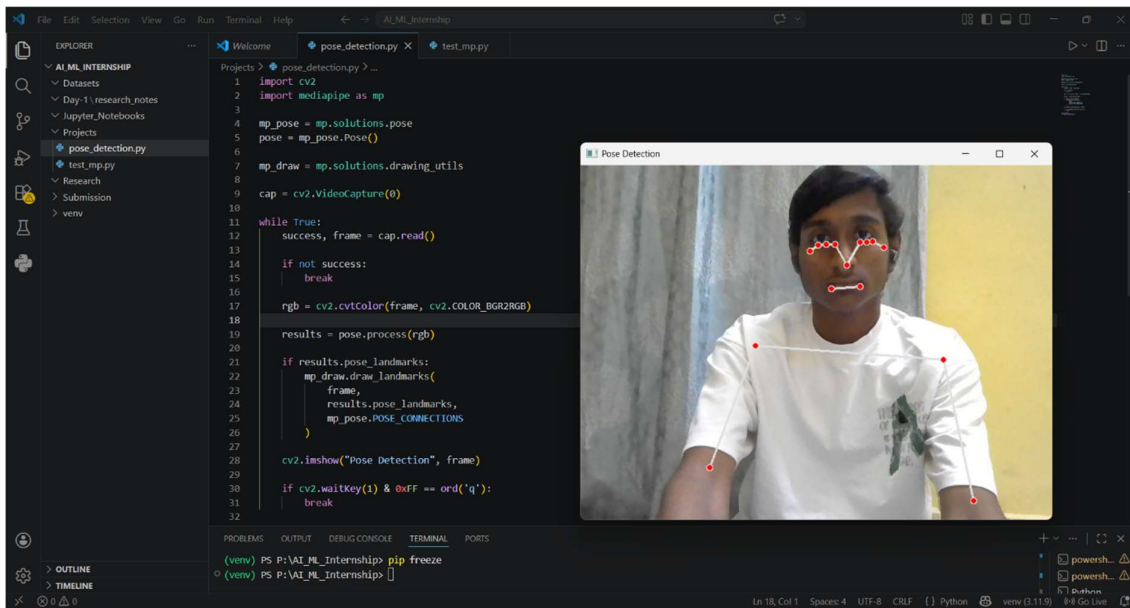
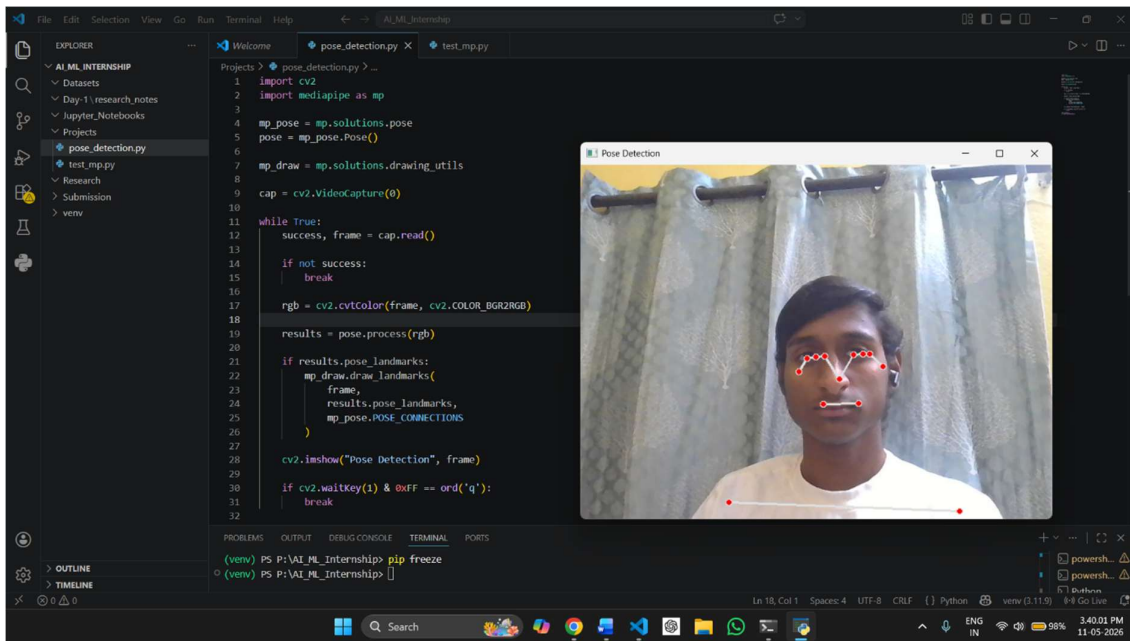
OUTPUT

Output and Results

The pose detection system successfully:

- opened the webcam
- detected body landmarks
- tracked body movement in real time

The output demonstrated practical implementation of computer vision and pose estimation techniques.



QUESTIONS ASKED IN TASK

What is Pose Detection?

Pose detection is a computer vision and artificial intelligence technique used to identify and track human body landmarks from images or videos.

The system detects key body points such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These body landmarks help AI systems understand:

- posture
- movement
- gestures
- physical activity

Pose detection works using machine learning algorithms and deep learning models that analyse visual data captured from cameras.

The technology is capable of performing real-time body tracking using webcams or mobile cameras.

Pose detection is widely used in:

- fitness applications
- healthcare systems
- sports analytics
- gaming
- motion tracking
- gesture recognition
- virtual reality
- fashion technology

In this internship task, pose detection was implemented using Python, OpenCV, and MediaPipe. The webcam successfully tracked body movement and displayed body landmark connections in real time.

Pose detection is an important area in AI and computer vision because it enables machines to visually understand human movement and interaction.

How Can AI Help Fashion Shopping?

Artificial Intelligence significantly improves fashion shopping by making online shopping more personalized, accurate, and interactive.

AI technologies help customers by:

- recommending suitable clothing
- predicting clothing sizes
- analysing fashion preferences
- enabling virtual try-on systems
- estimating body measurements

One of the biggest challenges in online fashion shopping is incorrect size selection. AI solves this problem using body measurement estimation and recommendation systems.

AI can analyse:

- customer preferences
- purchase history
- body shape
- fashion trends

and provide personalized product suggestions.

Applications of AI in fashion shopping include:

- virtual fitting rooms
- AI fashion assistants
- smart recommendations
- style analysis
- personalized advertisements

- inventory prediction
- trend forecasting

Virtual try-on systems allow users to digitally wear clothes using augmented reality and computer vision technologies before purchasing products.

Benefits of AI in fashion shopping:

- improved customer experience
- reduced return rates
- faster product discovery
- personalized recommendations
- increased shopping convenience
- better customer satisfaction

AI is transforming the fashion industry by making online shopping more intelligent, interactive, and efficient.

LEARNING OUTCOMES

During Day 1, these are the skills and concepts that I have learned:

- Python environment setup
- VS Code configuration
- Python package management
- OpenCV basics
- MediaPipe basics
- Pose detection systems
- AI applications in fashion technology
- Real-time webcam processing
- Debugging and troubleshooting AI environments

CONCLUSION

Day 1 provided practical exposure to AI/ML environment setup and computer vision technologies. A real-time pose detection system was successfully implemented using OpenCV and MediaPipe, and research was conducted on AI applications in fashion and virtual try-on systems. The session established a strong foundation for future AI/ML internship tasks and projects.